

REVIEW

Open Access



Simplicial complexes in network intrusion profiling: pattern construction through simplicial centralities

Mandala von Westenholz^{1,4*}, Martin Atzmueller^{2,3,4} and Tim Römer^{1,4}

*Correspondence:
Mandala von Westenholz
mvonwestenho@uni-osnabrueck.
de

Full list of author information is
available at the end of the article

Abstract

For studying intrusion detection data we consider data points referring to individual IP addresses and their connections. We build networks represented by graphs associated with those data points, such that vertices in a graph are constructed to denote the respective IP addresses, with the key property that attacked data points are part of the structure of the network. More precisely, this paper proposes a novel approach using simplicial complexes to model the desired network and the respective intrusions in terms of simplicial attributes, thus generalizing previous graph-based approaches. Applying adapted network centrality measures related to simplicial complexes yields patterns associated to vertices, which themselves contain a set of features. These are used to describe the attacked or the attacker vertices, respectively. Comparing this new strategy with classical concepts demonstrates the advantages of the presented approach using simplicial features for detecting and characterizing intrusions.

Keywords Simplicial complex, Network intrusion profiling, Simplicial patterns

Introduction

Network intrusion detection (see, e.g., (Mukherjee et al. 1994; Sommer and Paxson 2010)) is a prominent and important research direction due to growing challenges in cyber security, e.g., relating to the risk for all that personal data will be assaulted (see, e.g., (Rosenberg 2017; Walters and Novak 2021)), as well as increased cyber security measures in general (Tirumala et al. 2019). Hence, it is of particular interest to enhance our understanding for making sense of such situations and data, e.g., by identifying and considering groups of special IP addresses, like the ones that are attacked as well as the attackers themselves during a series of intrusion assaults.

For that, we apply graph-based methods. In general, graph-based approaches have found various applications in computer science, mathematics, and neighbouring disciplines (see, e.g., (Marisca et al. 2024; Kosk 2024; Tang 2023)). In Atzmueller et al. (2023), such an idea was used to model network intrusion detection data, in order to perform group-based analysis of critical situations, i.e., specific network intrusion events. More

precisely, IP addresses correspond to vertices of an associated directed graph modeled using the network intrusion data. In this graph, there is an edge between two vertices whenever there exists any kind of data exchange between the respective IP addresses. Given such a graph, the authors in Atzmueller et al. (2023) aimed to find patterns associated to vertices, which are themselves sets of well-chosen features of the IP addresses and the respective nodes. The goal is to find common properties (i.e., sets of features) of groups of vertices that best characterize groups of IP addresses of interest as good as possible, such as the ones related to attackers or attacked data points. For this purpose, graph centrality measures are used to construct specific features as properties of interesting subsets of the vertices of the graph.

It turns out that graph-based patterns described in Atzmueller et al. (2023) describe critical situations of IP addresses much better than non-graph-based patterns related to classical constructions. However, there is room for improvement, since graphs only enable the detection of possible pairwise interactions between IP addresses. In contrast, higher order structures, like the interaction of three or more addresses simultaneously are then not directly accessible. There is the need for generalizing the graph-based approach from above in a suitable way, which still is simple enough for practical application. This need is motivated by the latter observation.

Simplicial complexes are fulfilling this aim, and that is why we propose these for modeling, analysis and interpretation. Recall that given a set of vertices V , such a simplicial complex is a set of subsets of V , called *faces*, which is closed under taking subsets. Geometrically, each of these subset corresponds to the vertices of a simplex in a given real vector space.

In this work, we usually use *Vietoris–Rips complexes* (see, e.g., in general (Edelsbrunner and Harer 2022; Gilbert 1961; Hug and Reitzner 2016; Penrose 2003) and for specific contexts, e.g., (Akinwande and Reitzner 2020; Reitzner et al. 2024; Grygierek et al. 2020) where $V \subset \mathbb{R}^n$ and a set of vertices induces a face if and only if pairwise each two vertices are close enough with respect to a given metric. Given such a simplicial complex, we can associate patterns of features to its vertices using the simplicial structure of the complex. Such a simplicial approach turns out to be effective for describing and studying the structure of a network based on faces in various dimensions. For example, we might be interested in selecting a facet of the highest dimension containing a given vertex v as a feature of a pattern. Then, v is connected to the other vertices of this facet; in this way, we can then describe the connectivity of v concerning the whole network.

Observe that graphs appear exactly as 1-dimensional skeletons of simplicial complexes, only containing the faces of dimensions 0 and 1 – thus only providing a 1-dimensional perspective. In contrast, our new approach generalizes the methods in Atzmueller et al. (2023) using simplices in given complexes of higher dimensions. We focus on centrality measures related to simplicial complexes to yield new patterns associated to vertices, which are not visible in the world of graphs.

The rest of the paper is structured as follows. Section “[Preliminaries](#)” presents relevant basics on simplicial complexes. Next, Section “[Adjacencies and degrees for simplicial complexes](#)” provides suitable notions of adjacencies and degrees of faces of simplicial complexes. Note that there are several ways adjacencies can be defined (see (Serrano and Gómez 2020)). In this paper, we use the variant generalizing graph adjacencies. With these definitions, centralities for vertices are constructed in Section “[Simplicial](#)

centrality measures”, which are generalizations of graph-based centrality measures. In Section “**Building patterns**”, it is outlined how we can construct simplicial patterns with features, in particular, those ones relying on simplicial centrality measures. Next, purely graph-based patterns and simplicial ones are analyzed. For this, quality functions are introduced, which compare measures for vertices of interest (e.g., attackers in the network) in relation to all vertices, and measures for vertices of interest inside the support of patterns in relation to its complete support. It turns out that simplicial patterns based in high dimensional structures are significantly better for several complex networks than purely graph-based patterns, as studied in Atzmueller et al. (2023). We conclude the paper with an outlook in Section “**Outlook**”, presenting a discussion with respect to further work and interesting research directions how to investigate simplicial patterns as related to a network further.

Preliminaries

In this section, definitions and notation related to simplicial complexes are given, which are used all over the manuscript. For more details, for example, see Munkres (2018), Bobrowski and Kahle (2018), Kahle and Meckes (2013), Stanley (2007).

Definition 2.1 Let $V = \{v_1, \dots, v_m\}$ be a finite set. A family Δ of subsets of V is an *abstract simplicial complex*, if

for any $\sigma \in \Delta$ and $\tau \subseteq \sigma$ holds $\tau \in \Delta$.

For $0 \leq k \leq m - 1$, an element $\sigma \in \Delta$ with $|\sigma| = k + 1$ is called an *abstract k -simplex* and k is called the *dimension* of σ .

The highest possible dimension k of an abstract k -simplex in an abstract simplicial complex is called the *dimension* of the complex. Observe that every abstract simplicial complex can be realized as a *geometric simplicial complex* (see, e.g., (Munkres 2018)) in $\mathbb{R}^n$ with a sufficiently large n . In the following we are always considering abstract simplicial complexes. For brevity, we often skip the word “abstract” for the corresponding objects.

Example 2.2 The (abstract) simplicial complex

$$\{\emptyset, \{x_1\}, \{x_2\}, \{x_3\}, \{x_4\}, \{x_5\}, \{x_1, x_2\}, \{x_1, x_3\}, \{x_2, x_3\}, \\ \{x_4, x_5\}, \{x_2, x_5\}, \{x_2, x_4\}, \{x_1, x_2, x_3\}, \{x_4, x_5, x_2\}\}$$

can be realized in $\mathbb{R}^2$. A possible realization is illustrated in Fig. 1. Though interesting questions in research are related to such embeddings, they are not discussed in this manuscript, except implicitly when illustrating complexes in pictures.

Remark 2.3 For this work, it is important to note that simplicial complexes are very good models for describing networks. More precisely, objects in a network correspond to vertices. Such vertices are connected by an edge if the related objects fulfill a given relationship (which is based on the context of the network). A k -simplex with $k > 1$ in the simplicial complex specifies whether a set of at least $k + 1$ vertices fulfills some kind of pairwise relationship of interest.

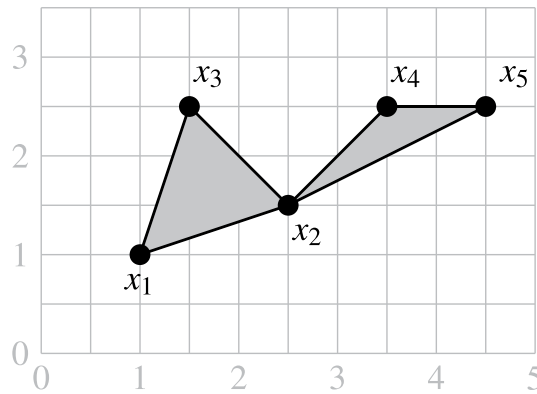


Fig. 1 Geometric realization of a simplicial complex

There are several ways to build a simplicial complex on a set of vertices. A key construction in various areas (see, e.g., (Edelsbrunner and Harer 2022; Gilbert 1961; Hug and Reitzner 2016; Penrose 2003)) is given in the following definition.

Definition 2.4 Let $V = \{v_1, \dots, v_m\} \in \mathbb{R}^n$, $r \in \mathbb{R}_{>0}$, and a given metric d on $\mathbb{R}^n$. The Vietoris–Rips complex $\mathcal{R}(V, r)$ is the simplicial complex, whose k -simplices $\sigma^{(k)}$ are the sets $\{v_{i_1}, \dots, v_{i_{k+1}}\} \subseteq V$ with

$$d(v_{i_l}, v_{i_j}) \leq r \text{ for all } j, l \in \{1, \dots, k+1\}.$$

The Vietoris–Rips complex is the *clique complex* (see, e.g., (Kozlov 2008)) of the underlying graph G consisting of its 0- and 1-dimensional faces. This means that whenever there exists a $(k+1)$ -clique in G , the corresponding k -simplex belongs to the simplicial complex. Note that there are many other ways to create simplicial complexes, like *C ech complexes*.

The Vietoris–Rips construction is the only one that is used in this paper. To construct a Vietoris–Rips complex, it is required to define the distance between two data points $x_1, x_2 \in \mathbb{R}^n$ via a metric d . There are several reasonable options for that. Below, there are some suggestions, where the objects according to Remark 2.3 are IP addresses and one is interested in their interaction in the internet.

- Euclidean distance $d_1(x_1, x_2) = \|x_2 - x_1\|_2$
- Other spatial distances like $d_2(x_1, x_2) = \|x_2 - x_1\|_\infty$ or $d_3(x_1, x_2) = \|x_2 - x_1\|_1$
- $d_4(x_1, x_2) = \begin{cases} \frac{1}{2} & \text{if there is a possible data exchange of size } \epsilon \text{ between } x_1, x_2 \\ \frac{\epsilon}{2} & \text{otherwise} \end{cases}$
- $d_5(x_1, x_2) = \begin{cases} 0 & \text{if there is a possible data exchange between } x_1 \text{ and } x_2 \\ 1 & \text{otherwise} \end{cases}$

Example 2.5 The simplicial complex in Example 2.2 is a Vietoris–Rips complex with $V = \{x_1, x_2, x_3, x_4, x_5\}$, $d = d_1$, and $r = 3$.

If one is interested in relationships and in activities that are taking place at a special location, e.g., in a company, it might be helpful to consider spatial distances, like the Euclidean one. There are situations where it might be more reasonable to use distances which are more closely related to possible data exchange, like d_4 and d_5 .

Recall that the constructions of graphs in Atzmueller et al. (2023) are based on placing an (undirected) edge between two vertices v and w in a finite vertex set V corresponding

to IP-addresses, when they interact (in one or two directions). These are exactly Vietoris–Rips graphs (which are also called *Gilbert graphs* or *random geometric graphs* Gilbert (1961)) using the metric d_5 .

Example 2.6 Let us reconsider the simplicial complex of Example 2.2. Some exemplary distances are given by $d_1, \dots, d_5$ from above, which are

$$d_1(x_1, x_2) = \frac{\sqrt{10}}{4}, d_2(x_1, x_2) = \frac{3}{2}, d_3(x_1, x_2) = 2, d_4(x_1, x_2) \text{ depends on } e, d_5(x_1, x_2) = 0.$$

Note that one can also distinguish between incoming and outgoing data by adding directions to our constructions so far. In this work, we always consider undirected situations.

Adjacencies and degrees for simplicial complexes

Notions like adjacencies and degrees in graphs are well known (see, e.g., (Diestel 2025)). Generalizing this to simplicial complexes and based on the work in Serrano and Gómez (2020), the main goal of this section is to introduce various types of *adjacencies* and *degrees* in such complexes. Later, these notions are then used, in particular, to construct *features* of vertices, which use simplicial centrality measures.

From now on a simplicial complex is always denoted by Δ and $\sigma^{(q)}$ is a simplex in Δ of dimension q . First we give the following definitions from Serrano and Gómez (2020, Section 2.1) which will be important in the upcoming parts of this paper.

Definition 3.1 Let $\sigma^{(q)} \neq \emptyset$ and $\sigma'^{(q')} \neq \emptyset$ be different simplices in Δ and $p \in \mathbb{N}$ with $1 \leq p \leq \dim \Delta$. Then, we define the following:

- (i) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *p-upper adjacent*, denoted by

$$\sigma^{(q)} \sim_{U_p} \sigma'^{(q')}$$

if there exists a p -simplex $\tau^{(p)} \in \Delta$, having both $\sigma^{(q)}$ and $\sigma'^{(q')}$ as faces.

- (ii) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *strictly p-upper adjacent*, denoted by $\sigma^{(q)} \sim_{U_p^*} \sigma'^{(q')}$, if

$$\sigma^{(q)} \sim_{U_p} \sigma'^{(q')} \text{ and } \sigma^{(q)} \not\sim_{U_{p+1}} \sigma'^{(q')}.$$

- (iii) The *p-upper degree* of $\sigma^{(q)}$ is

$$\deg_U^p(\sigma^{(q)}) = |\{\sigma''^{(q'')} \in \Delta \mid \sigma^{(q)} \sim_{U_p} \sigma''^{(q'')}\}|.$$

- (iv) The *(h, p)-upper degree* of $\sigma_1^{(q)}$ is

$$\deg_U^{(h,p)}(\sigma^{(q)}) = |\{\sigma''^{(q+h)} \in \Delta \mid \sigma''^{(q+h)} \sim_{U_p} \sigma^{(q)}\}|.$$

- (v) The *strict (h, p)-upper degree* of $\sigma^{(q)}$ is

$$\deg_U^{(h,p)*}(\sigma^{(q)}) = |\{\sigma''^{(q+h)} \in \Delta \mid \sigma''^{(q+h)} \sim_{U_{p*}} \sigma^{(q)}\}|.$$

(vi) The *maximal simplicial upper degree* of $\sigma^{(q)}$ is

$$\deg_U^*(\sigma^{(q)}) = \sum_{h=1}^{\dim \Delta - q} \deg_U^{(h, (q+h))^*}(\sigma^{(q)}).$$

Notice that the maximal simplicial upper degree $\deg_U^*(\sigma^{(q)})$ counts the number of maximal simplices, which have dimension strictly higher than q , that contain the simplex $\sigma^{(q)}$.

Example 3.2 (i) Two (abstract) simplices, which are faces of a solid tetrahedron are 3-upper adjacent.

(ii) Two vertices which share a common edge are 1-upper adjacent; they are even strict 1-upper adjacent, if they are not lying in a common triangle.

There are also other variations of the concept of adjacency and degree, which are also introduced in Serrano and Gómez (2020, Section 2.1), that are listed below.

Definition 3.3 Let $\sigma^{(q)} \neq \emptyset$ and $\sigma'^{(q')} \neq \emptyset$ be different simplices in Δ and $p \in \mathbb{N}$ with $0 \leq p \leq \dim \Delta - 1$. Then, we define the following:

(i) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *p-lower adjacent*, if $|\sigma^{(q)} \cap \sigma'^{(q')}| \geq p + 1$. This is denoted by

$$\sigma^{(q)} \sim_{L_p} \sigma'^{(q')}.$$

(ii) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *strictly p-lower adjacent*, if $|\sigma^{(q)} \cap \sigma'^{(q')}| = p + 1$. This is denoted by

$$\sigma^{(q)} \sim_{L_{p*}} \sigma'^{(q')}.$$

(iii) The *p-lower degree* of $\sigma^{(q)}$ is

$$\deg_L^p(\sigma^{(q)}) = |\{\sigma''^{(q'')} \in \Delta \mid \sigma^{(q)} \sim_{L_p} \sigma''^{(q'')}\}|.$$

(iv) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *p-adjacent*, if they are strictly p-lower adjacent and not p' -upper adjacent for $p' = q + q' - p$. This is denoted by

$$\sigma^{(q)} \sim_{A_p} \sigma'^{(q')}.$$

(v) The *p-adjacency degree* of $\sigma^{(q)}$ is

$$\deg_A^p(\sigma^{(q)}) = |\{\sigma''^{(q'')} \in \Delta \mid \sigma^{(q)} \sim_{A_p} \sigma''^{(q'')}\}|.$$

(vi) $\sigma^{(q)}$ and $\sigma'^{(q')}$ are called *maximal p-adjacent*, denoted by

$$\sigma^{(q)} \sim_{A_p^*} \sigma'^{(q')}$$

if and only if $\sigma^{(q)} \sim_{A_p} \sigma'^{(q')}$ and $\sigma'^{(q')} \not\sim_{A_p} \sigma''^{(q'')}$ whenever $\sigma^{(q)} \sim_{A_p} \sigma''^{(q'')}$.

(vii) The *maximal p-adjacency degree* of $\sigma^{(q)}$ is

$$\deg_A^{p*}(\sigma^{(q)}) = |\{\sigma''(q'') \in \Delta | \sigma^{(q)} \sim_{A_p^*} \sigma''(q'')\}|.$$

(viii) The *maximal simplicial degree* of $\sigma^{(q)}$ is

$$\deg^*(\sigma^{(q)}) = \deg_A^*(\sigma^{(q)}) + \deg_U^*(\sigma^{(q)}),$$

where

$$\deg_A^*(\sigma^{(q)}) = \sum_{p=0}^{q-1} \deg_A^{p*}(\sigma^{(q)}).$$

The following examples are some illustrations for the definitions introduced so far.

Example 3.4 Consider the simplicial complex from Example 2.2. The simplices x_1, x_2, x_3 and x_2, x_4, x_5 exhibit the properties outlined below:

- (i) x_2 is a vertex of both simplices. Hence, they are 0-lower adjacent. They are strictly 0-lower adjacent as well, as they share no simplex of higher dimension.
- (ii) Both simplices are not included in a common simplex of higher dimension, so they are not p -upper adjacent for any p .
- (iii) They are 0-adjacent, since they are strictly 0-lower adjacent and they are not 4-upper adjacent.
- (iv) The 0-adjacency degree of $\{x_1, x_2, x_3\}$ is 3, since besides $\{x_2, x_4, x_5\}$, also $\{x_2, x_4\}$ and $\{x_2, x_5\}$ are 0-adjacent to $\{x_1, x_2, x_3\}$.

In the upcoming sections, we will focus on those situations, where one simplex is always a vertex, although Definition 3.3 is of course also usable for simplices of higher dimensions.

In this special case we observe for graphs, that two vertices are called adjacent, when they both share a common edge. This graph-adjacency is covered by the definitions of adjacencies in simplicial complexes above as well. More precisely, we have the following result.

Lemma 3.5 Let $G = (V, E)$ be a graph and $v \in V$ be a vertex. Then, the following properties hold:

- (i) Let $w \in V$ be another vertex in G . Then, w and v are graph-adjacent if and only if they are 1-upper adjacent.
- (ii) If $v \neq w \in V$, then v, w are not p -lower adjacent and they are not p -adjacent for any $0 \leq p \leq \dim \Delta - 1$.

Proof (i) If v and w are graph-adjacent, then it follows immediately from the definitions, that they are 1-upper adjacent. On the other hand, if v and w are 1-upper adjacent, then they are both a face of a 1-simplex (i.e., an edge) of G . (ii) The vertices v and w have no non-trivial face in common and are not p -lower adjacent for any possible p . This implies that they are also not p -adjacent for such a p . $\square$

Lemma 3.6 Let $v \in \Delta$ be a vertex. Then

- (i) $\deg_L^p(v) = \deg_A^p(v) = 0$ for all $p > 0$,
- (ii) $\deg_L^p(v) = 0 = \deg_A^p(v)$ for $p = 0$,
- (iii) $\deg^*(v) = \deg_U^*(v)$.

Proof (i) Since a vertex has no other non-trivial simplices as subsets, the conclusion follows.

(ii) By definition v is not 0-lower adjacent to itself. Hence, $\deg_L^p(v) = 0 = \deg_A^p(v) = 0$.

(iii) Since $\deg_A^p(v) = 0$, we have $\deg_A^{p*}(\sigma^{(q)}) = 0$.

This concludes the proof. $\square$

Below, we are mainly interested in simplices in which a given vertex is included. In this context, only p -upper adjacencies and related degrees are of interest. The upper-adjacency is also reasonable to use, as it is a generalization of graph-adjacencies due to Lemma 3.5. Hence, in the following we only use p -upper adjacencies.

Simplicial centrality measures

In this section, we introduce centrality measures for simplicial complexes which are motivated by the same concept related to graphs. In the following we focus on the degree and closeness centralities. These generalizations of well-known graph degree and closeness centralities (see, e.g., (Borgatti and Everett 2006; Newman 2018)) have all been proposed in Serrano and Gómez (2020). With these centrality measures we create features for vertices of a complex as announced above in the introduction in Section “Introduction”. To be more formal, a *centrality measure* is a function

$$c : I \rightarrow \mathbb{R},$$

where I is the set, which is usually called the *set of individuals*. If I is a simplicial complex, then a more important simplex $\sigma \in I$, i.e., one with a significantly high *centrality* $c(\sigma)$, contains way more valuable information regarding the considered network than simplices with smaller centralities. There are many possibilities to define such a centrality measure. Hereby, the notation c_σ is an abbreviation for the function value $c(\sigma)$.

Motivated by definitions and suggestions in Serrano and Gómez (2020), we introduce the following centrality measures for simplicial complexes. Observe that in that work, the authors propose centrality measures with images in $[0, 1]$. But, to be consistent with the considered centrality measures in Atzmueller et al. (2023), such normalizations are neglected in this manuscript. Then, centrality measures in Atzmueller et al. (2023) are special cases of the simplicial versions given below. For simplicity, the following concepts are only defined for vertices of simplicial complexes, since, in this paper in considered applications individuals are always vertices. As already mentioned, in the end of Section “Adjacencies and degrees for simplicial complexes”, only upper degrees are considered. The first centrality measure of importance is the one from the following definition.

Definition 4.1 Let $\sigma \in \Delta$ be a vertex and $p \in \mathbb{N}_{>0}$. Then

$$c_\sigma^{D^p} = \deg_U^p(\sigma)$$

is the p -degree centrality of σ and

$$c_{\sigma}^{D^*} = \deg^*(\sigma)$$

is the *maximal simplicial degree centrality* of σ .

See (Serrano and Gómez 2020, Def. 12, 13) for a related definition. An important property of this centrality measure is discussed in the following theorem.

Theorem 4.2 The p -degree centrality of a vertex $\sigma \in \Delta$ fulfills

$$c_{\sigma}^{D^p} \leq \sum_{j=1}^{p+1} \left(\deg_U^{(0,1)}(\sigma) + 1 \right) - 1. \quad (1)$$

Proof Observe that $(j-1)$ -dimensional faces in Δ with $2 \leq j \leq p+1$ are p -upper adjacent to σ , if they lie in a common p -dimensional simplex with σ . Given such a face, all its vertices not equal to σ have the property that they induce an edge in Δ together with σ , i.e., they are $(0, 1)$ -adjacent to σ . Note that σ might be a vertex of such a face or not. Hence, it suffices to count all possible $(j-1)$ -faces induced by vertices which are either σ or $(0, 1)$ -adjacent to σ , accepting an over-count of such situations. Finally, there are at most $\deg_U^{(0,1)}(\sigma)$ many 0-dimensional faces in Δ which are p -upper adjacent to σ . This concludes the proof. $\square$

There is some significant computational effort required to provide an upper bound for the p -degree centrality of σ through Theorem 4.2. Next, we define a further degree centrality which allows alternative upper bounds.

Definition 4.3 Let $\sigma \in \Delta$ be a vertex and $p \in \mathbb{N}_{>0}$. Then

$$c_{\sigma}^{D^{(p,p)}} = \deg_U^{(p,p)}(\sigma)$$

is the (p, p) -degree centrality of σ and

$$c_{\sigma}^{D^{(p,p)*}} = \deg_U^{(p,p)*}(\sigma)$$

is the (p, p) -strict degree centrality of σ .

See (Serrano and Gómez 2020, Def. 10, 11) for a related definition. Note that the (p, p) -degree centrality of σ is the number of p -dimensional faces of Δ that contain σ , and the (p, p) -strict degree centrality is the corresponding strict version. Since σ is always a simplex of dimension 0, we have

$$c_{\sigma}^{D^*} = \sum_{p=1}^{\dim \Delta} \deg_U^{(p,p)*}(\sigma). \quad (2)$$

Remark 4.4 A new upper bound for $c_{\sigma}^{D^p}$ is

$$c_{\sigma}^{D^p} \leq \deg_U^{(p,p)}(\sigma) \cdot (2^{p+1} - 2) = c_{\sigma}^{D^{(p,p)}} \cdot (2^{p+1} - 2). \quad (3)$$

This bound follows from the fact that every non-trivial face, except for σ , of a p -simplex, containing σ , is p -upper adjacent to σ .

The upper bound in (3) is often better than the bound in Theorem 4.2. For example, consider a simplicial complex Δ which has exactly one tetrahedron that contains σ , and

σ has additionally t many 1-upper adjacent vertices, which are not pairwise connected with each other by an edge. Then, for $p = 3$, the bound in (3) gives 14, independent of t , where the other one is a growing function in t .

The disadvantage of the bound in (3) in comparison to the bound in (1) is the need to find all p -simplices in Δ , which contain σ , which is usually of more computational effort especially for $p \gg 0$, compared to determine all $(0, 1)$ -adjacent vertices of σ .

Lemma 4.5 The (p, p) -degree centrality of a vertex $\sigma \in \Delta$ is anti-monotonic, i.e.

$$c_{\sigma}^{D^{(p,p)}} \leq c_{\sigma}^{D^{(q,q)}} \text{ for } 1 \leq q \leq p.$$

Proof For every p -simplex τ that contains σ there exists a lower-dimensional q -face $\tilde{\sigma}$ of τ for each $q = p - k$ with $0 \leq k \leq p - 1$ that contains σ as well. $\square$

An upper bound for the maximal simplicial degree centrality is given in the following theorem.

Theorem 4.6 Let $\sigma \in \Delta$ be a vertex. For the maximal degree centrality of σ we have

$$c_{\sigma}^{D^*} \leq c_{\sigma}^{D^{(1,1)}} \cdot \dim \Delta. \quad (4)$$

Proof The assertion follows from the next equation.

$$\begin{aligned} c_{\sigma}^{D^*} &= \sum_{p=1}^{\dim \Delta} \deg_U^{(p,p)*}(\sigma) \leq \sum_{p=1}^{\dim \Delta} \deg_U^{(p,p)}(\sigma) \\ &\leq \dim \Delta \cdot \max_{p=1, \dots, \dim \Delta} \deg_U^{(p,p)}(\sigma) \\ &\leq \dim \Delta \cdot \deg_U^{(1,1)}(\sigma) = \dim \Delta \cdot c_{\sigma}^{D^{(1,1)}}. \end{aligned}$$

The first inequality follows directly from the definitions and the second one is trivial to see. The third inequality is a consequence of Lemma 4.5 and the equation is just the definition of the (p, p) -degree centrality. $\square$

Next, we discuss a typical situation where the maximal simplicial degree centrality $c_{\sigma}^{D^*}$ of a vertex $\sigma \in V$ is of interest. Consider the individual σ_2 , i.e., the corresponding vertex, in the network induced from Fig. 2 shown in Example 4.7 below. It has a higher maximal simplicial degree centrality $c_{\sigma_2}^{D^*} = 6$ than the individual σ_1 with $c_{\sigma_1}^{D^*} = 1$ (see below). Given the context of a potential application, there might be a higher risk of σ_2 to be intruded, because the intrusion of σ_2 has a higher impact with respect to aspects measured by $c_{\sigma}^{D^*}$.

Example 4.7 also illustrates why it might not be sufficient to consider only the (p, p) -degree centrality for the highest p as a feature to evaluate the importance of a vertex. If we take the maximal simplicial degree centrality as well into account as a feature in the discussed situation, then we obtain an indicator, if there may be a lot of simplices, which are not of maximal dimension, that include v as a vertex.

Example 4.7 In the following illustration we depict a simplicial complex in $\mathbb{R}^3$ which has solid simplices and tetrahedrons. The vertices σ_1, σ_2 have (p, p) -degree centralities

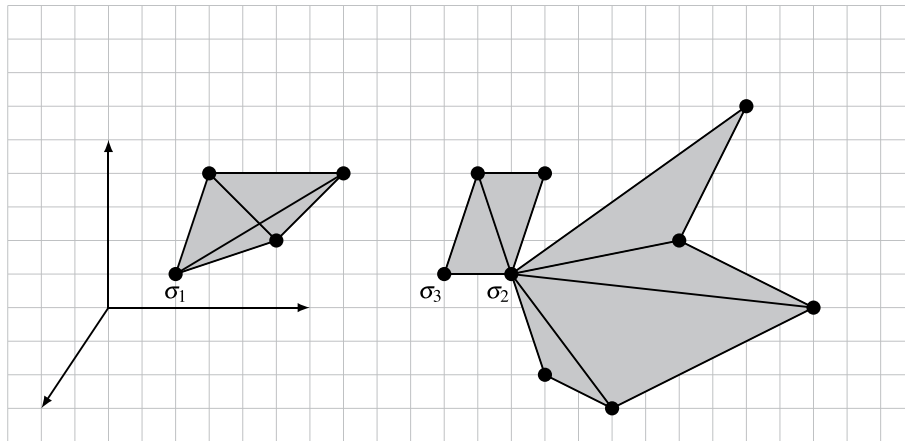


Fig. 2 Two simplicial complexes realized in $\mathbb{R}^3$.

$$c_{\sigma_1}^{D^{(3,3)*}} = 1, c_{\sigma_2}^{D^{(3,3)*}} = 0 \text{ and } c_{\sigma_1}^{D^{(2,2)*}} = 0, c_{\sigma_2}^{D^{(2,2)*}} = 6.$$

Moreover,

$$c_{\sigma_1}^{D^*} = 1 < c_{\sigma_2}^{D^*} = 6,$$

although for the maximal p , the (p, p) -degree centrality of σ_1 is higher than the one of σ_2 . Furthermore, the upper bounds from Eq. (4) are

$$c_{\sigma_1}^{D^*} \leq 3 \cdot c_{\sigma_1}^{D^{(1,1)}} = 9 \text{ and } c_{\sigma_2}^{D^*} \leq 2 \cdot c_{\sigma_1}^{D^{(1,1)}} = 12.$$

The authors of Serrano and Gómez (2020) discuss suggestions how to generalize *closeness centralities* for vertices in graphs to new ones for simplices in simplicial complexes. Their proposals are based on the notation of p -adjacency. Since, in this article, we only use p -upper adjacency, we will work with the following modified versions of the definitions in Serrano and Gómez (2020, Sections 3.2, 3.3).

Definition 4.8 Let Δ be a simplicial complex and $p \in \mathbb{N}_{>0}$. Let $\sigma, \tilde{\sigma} \in \Delta$ be two vertices. $\sigma, \tilde{\sigma}$ are called p -connected, if there is a path in the underlying graph of Δ starting in σ , ending in $\tilde{\sigma}$, and where two consecutive vertices of the path are p -upper adjacent in Δ . The simplicial complex Δ is called p -upper connected if each pair of two vertices $\sigma, \tilde{\sigma} \in \Delta$ is p -connected.

Definition 4.9 Let $\sigma, \tilde{\sigma} \in \Delta$ be two vertices and $p \in \mathbb{N}_{>0}$.

(i) A p -walk between σ and $\tilde{\sigma}$ of length $r \in \mathbb{N}_{\geq 1}$ is a sequence

$$\sigma_1 \tau_1^{(p_1)} \sigma_2 \tau_2^{(p_2)} \dots \tau_{r-1}^{(p_{r-1})} \sigma_r$$

such that $\sigma = \sigma_1, \sigma_r = \tilde{\sigma}, \sigma_i$ and σ_{i+1} are p_i -upper adjacent with respect to $\tau_i^{(p_i)}$ for $i \in \{1, \dots, r-1\}$ and $p = \max\{p_1, \dots, p_r\}$.

(ii) The p -distance $d_p(\sigma, \tilde{\sigma})$ between σ and $\tilde{\sigma}$ is the length of the shortest p -walk between σ and $\tilde{\sigma}$. If there are no p -walks, then the p -distance is $d_p(\sigma, \tilde{\sigma}) = \infty$.

The closeness centrality for vertices in graphs can now be generalized.

Definition 4.10 Let $\sigma \in \Delta$ be a vertex and $p \in \mathbb{N}_{>0}$. The p -closeness centrality of σ is

$$c_{\sigma}^{C^p} = \frac{1}{\sum_{\substack{\sigma \neq \tilde{\sigma} \in \Delta \\ \text{is a vertex}}} d_p(\sigma, \tilde{\sigma})}, \text{ where } \frac{1}{\infty} = 0 \text{ and}$$

$$c_{\sigma}^{C^*} = \sum_{1 \leq p \leq \dim \Delta} c_{\sigma}^{C^p}$$

is the *maximal closeness centrality* of σ .

Similar to the degree centrality, an upper bound for the maximal closeness centrality is given in the next theorem.

$$\text{Theorem 4.11 } c_{\sigma}^{C^*} \leq c_{\sigma}^{C^1} \cdot \dim \Delta. \quad (5)$$

Proof The assertion of the Theorem follows from Definition 4.10 and the claim that

$$c_{\sigma}^{C^1} \geq c_{\sigma}^{C^p} \text{ for all } 1 \leq p \leq \dim \Delta,$$

if there exist a p -walks from σ .

This is demonstrated as follows.

In case that two vertices σ and $\tilde{\sigma}$ are connected via a p -(or lower)-dimensional simplex, then they are also contained in a 1-face of that simplex, i.e., in an edge. Therefore, if there exists a p -walk between σ and $\tilde{\sigma}$, then there exists also a 1-walk, which has the same length as the p -walk, but there might also exist a 1-walk with a smaller length. It follows

$$d_p(\sigma, \tilde{\sigma}) \geq d_1(\sigma, \tilde{\sigma})$$

for all p . □

Note that this upper bound for $c_{\sigma}^{C^*}$ is similar to the bound $c_{\sigma}^{D^*}$ in Eq. (4).

Example 4.12 Recall that the illustration in Example 4.7 induces two connected simplicial complexes. Then, the closeness centralities in the right simplicial complex Δ are computed for the vertex σ_3 via

$$c_3^{C^1} = 2 \frac{1}{1} + 6 \frac{1}{2} = c_3^{C^2}.$$

If we add a new vertex by a new edge to σ_3 and, thus, construct a new simplicial complex Δ_1 , then we see that there is the following new result for $c_3^{C^1}$ in Δ_1 :

$$c_3^{C^1} = 3 \frac{1}{1} + 6 \frac{1}{2}.$$

But $c_3^{C^2}$ still equals to its old value. Hence, we see that

$$c_3^{C^1} \geq c_3^{C^2} \text{ in } \Delta_1.$$

Since there is no tetrahedron in Δ_1 , the maximal closeness centrality measure is given by

$$c_3^{C^*} = c_3^{C^2} + c_3^{C^1} < 2c_3^{C^1}.$$

As a conclusion we see that the upper bound in Theorem 4.11 might be strict.

For every centrality measure suggested in this section so far, there is a maximal centrality version, for which a computation for all possible p is necessary. Thus, it is of interest to decide whether this is really needed. It might be more reasonable to compute the relevant information just for some p which are of high interest, or to use upper bounds, like in Eqs. (4) and (5).

Next, we give a definition of *eigenvector centrality* for vertices in a simplicial complex Δ similar to the one in Serrano and Gómez (2020, Section 3.2). This shows, by way of example, that other known centrality measures for vertices in graphs can be generalized as well. But first, we make some preliminary considerations. *Irreducible* matrices are square matrices A that are not conjugated to an upper block triangular matrix $B = PAP^{-1}$ with two non-trivial blocks by using (only) permutation matrices P .

Lemma 4.13 Let $A = (a_{ij})$ be the $(m \times m)$ -adjacency matrix of a simple graph G . A is irreducible if and only if G is connected.

Proof A being irreducible is equivalent to the fact that there do not exist any two disjoint sets $I \neq \emptyset$ and $J \neq \emptyset$, which are subsets of $\{1, \dots, m\}$, with $I \cup J = \{1, \dots, m\}$, where for each pair $(i, j) \in I \times J$ it holds $a_{ij} = 0$. This is equivalent to G being a connected graph. $\square$

The next corollary is well-known for an $(m \times m)$ -non-negative matrix A and an associated directed graph $\mathcal{G}(A)$ (see, e.g., (Gaubert and Gunawardena 2004, Theorem 1)) and can be given in an undirected setting.

Corollary 4.14 Let G be a connected simple graph with at least one edge, A be its graph-adjacency matrix, and $\lambda > 0$ the largest eigenvalue of A . Then, A has a unique normalized eigenvector v with respect to λ that has non-negative entries.

Proof This follows by Lemma 4.13 and the version of *Perron–Frobenius* in Lemmens and Nussbaum (2012, Theorem 1.1.2 (i)), where $r(A)$ is the largest eigenvalue, since $A \neq 0$. $\square$ Now we are going to define eigenvector centrality for vertices $\sigma_1, \dots, \sigma_m$ (which are fixed for the remaining part of this section) by using the upcoming theorem (see (Serrano and Gómez 2020, Chapter 3.2)). For $1 \leq p \leq \dim \Delta$ the p -adjacency matrix A_p of Δ is defined via its entries

$$a_{ij} = \begin{cases} 0 & \text{if } i = j \text{ or } \sigma_i \not\sim_{U_p} \sigma_j, \\ 1 & \text{otherwise.} \end{cases}$$

Theorem 4.15 Let Δ be a p -upper connected simplicial complex with p -adjacency matrix $A_p \neq 0$. Let $\lambda > 0$ be the largest eigenvalue of A_p . Then, A_p has a unique normalized eigenvector c^{E^p} with respect to λ that has non-negative entries.

Proof The theorem follows directly from the fact that the p -adjacency matrix A_p of Δ is also the graph-adjacency matrix A of the underlying connected graph G of Δ (also called its 1-skeleton), where additionally all edges are removed that do not lie in a p -simplex. An application of Corollary 4.14 concludes the proof. $\square$

Now we are able to introduce the following definition.

Definition 4.16 Let $p \in \mathbb{N}_{>0}$ and Δ be a p -upper connected simplicial complex with $A_p \neq 0$ and let c^{E^p} be chosen as in Theorem 4.15. Set $c^{E^p} = (c_1^{E^p}, \dots, c_m^{E^p}) \in \mathbb{R}_{\geq 0}^m$. The p -eigenvector centrality is defined by mapping σ_i to

$$c_{\sigma_i}^{E^p} = c_i^{E^p}.$$

Moreover, the *maximal simplicial eigenvector centrality* is defined as

$$c_{\sigma_i}^{E^*} = \sum_{1 \leq p \leq \dim \Delta} c_{\sigma_i}^{E^p}.$$

Remark 4.17 The p -eigenvector centrality associates a score $c_{\sigma_i}^{E^p}$ to every vertex $\sigma_i \in \Delta$, which is derived in a particular way from $(A_p - \lambda I_m)x = 0$, where I_m is the $(m \times m)$ -identity matrix. More precisely, using the latter equation $c_{\sigma_i}^{E^p}$ can be computed as

$$c_{\sigma_i}^{E^p} = \frac{1}{\lambda} \sum_{j=1}^m a_{ij} c_{\sigma_j}^{E^p}.$$

Note that the p -eigenvector centrality from Definition 4.16 for a p -upper connected simplicial complex is indeed a generalization of the eigenvector centrality for connected graphs (see, e.g., (Borgatti and Everett 2006; Newman 2018)) for $p = 1$.

The next example shows that the maximal simplicial eigenvector centrality is in general not monotone as a function in p .

Example 4.18 We claim that in the following 2-connected simplicial complex Δ from Fig. 3 there exist two vertices $\sigma, \tilde{\sigma} \in \Delta$ with

$$c_{\sigma}^{E^1} > c_{\sigma}^{E^2} \text{ and } c_{\tilde{\sigma}}^{E^2} > c_{\tilde{\sigma}}^{E^1}.$$

Indeed, the adjacency matrices A_1 for $p = 1$ and A_2 for $p = 2$ are

$$A_1 = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \text{ and } A_2 = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}.$$

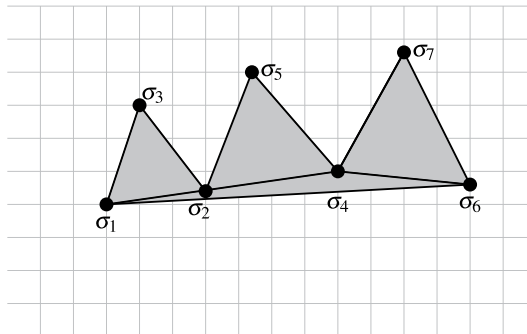


Fig. 3 Simplicial complex Δ realized in $\mathbb{R}^2$.

They just differ by the entries (1, 6) and (6, 1). For A_1 let v_1 be the normalized eigenvector for the largest eigenvalue and observe that

$$v_1 \approx (1.33, 1.70, 1, 1.70, 1.12, 1.33, 1).$$

Choosing similar v_2 for A_2 we obtain

$$v_2 \approx (1, 1.81, 1, 1.81, 1.29, 1, 1).$$

Hence, we finally find

$$c_6^{E^1} > c_6^{E^2} \text{ and } c_4^{E^2} > c_4^{E^1}.$$

Building patterns

In this section, we aim to find sets of suitable *features*, which describe certain properties of interest of vertices in given simplicial complexes, as attackers or attacked ones. Then, *patterns* are represented as exactly these sets of features. We give strategies and possibilities for constructing such patterns. Our definitions are motivated by the ones given in Atzmueller et al. (2023, 2019). In comparison to Atzmueller et al. (2023), the patterns will not only be equipped with graph-based features, but also with simplicial ones with respect to Vietoris–Rips complexes. In particular, centrality measures are taken into account (see Section “[Simplicial centrality measures](#)”). In the following formal scenario, we always consider I as the set of individuals where the individuals $i \in I$ own certain properties.

Definition 5.1 A *feature* is a function

$$f : I \rightarrow P_f,$$

where P_f is the set of properties depending on f . Moreover the *property function* f_b is defined via the following indicator function for a *property* $b \in P_f$.

$$f_b(i) := \mathbb{1}(i \in f^{-1}(\{b\})).$$

This means that a feature maps an individual $i \in I$ to its property $b \in P_f$, where P_f might be chosen differently for each feature. The function f_b indicates whether $i \in I$ fulfills b . Moreover we consider in the following the set F , which is the set of features, as well as $\mathcal{F} = \{f_b \mid f \in F \text{ and } b \in P_f\}$, which is the set of property functions.

An *attributed* simplicial complex Δ is a simplicial complex where each vertex $\sigma \in \Delta$ has an associated *dataset*

$$D(\sigma) = (i, F(i)),$$

where $i \in I$ is chosen and $F(i) = \{f(i) \mid f \in F\}$, in which the information regarding the considered features about the corresponding individual i to σ is stored.

From now on only attributed simplicial complexes are considered and F as above is fixed. In this manuscript we build a pattern by using features.

Definition 5.2 Let $f^1, \dots, f^k \in F$ and choose $b_j \in P_{f^j}$ for $j = 1, \dots, k$. A set

$$p = \{f_{b_1}^1, \dots, f_{b_k}^k\}$$

is called a k -*pattern* with respect to F . We denote the set of k -patterns by $\mathcal{F}^{(k)}$. Then, we say that $p \in \mathcal{F}^{(k)}$ is *true* with respect to $i \in I$, if

$$p(i) = \{f_{b_1}^1(i), \dots, f_{b_k}^k(i)\} = \{1\}.$$

The *support* of p is

$$s_p = \{i \in I \mid p(i) = \{1\}\} \subseteq I,$$

which is also called the p -*fulfilling individuals*. Set $i_p := |s_p|$.

Remark 5.3 (i) Let Δ be an attributed simplicial complex and p be a compatible pattern.

This pattern can also be considered as true or false with respect to a vertex $\sigma \in \Delta$ by taking the value $p(i)$, where i is the first component of $D(\sigma)$.

(ii) In the literature, e.g., in Atzmueller et al. (2023), there exist various ways of defining patterns. For example, one could equip the patterns with more structure by using tuples instead of sets. This provides the opportunity to consider multiplicities and to order the features. In this paper, patterns are always sets, which is suitable for the considered applications below.

For binary functions as already considered above, it is useful to introduce the following notation.

Definition 5.4 Let p be a pattern. A *target* t is a binary function $t : I \rightarrow \{0, 1\}$. The *target share* of t with respect to p is

$$t_p := \frac{|\{i \in s_p \mid t(i) = 1\}|}{i_p} = \frac{|\{i \in I \mid p(i) = 1 \text{ and } t(i) = 1\}|}{i_p}.$$

For a given target, we search for patterns with a high target share. To find a useful measurement for the pattern quality and, thus, for its interestingness, we consider a *quality function* (see (Atzmueller 2015; Atzmueller et al. 2023; Grosskreutz et al. 2008) for further details).

Definition 5.5 Given a target t , a k -*quality function* is a real-valued function

$$q_t : \mathcal{F}^{(k)} \rightarrow \mathbb{R}.$$

The *quality* of a k -pattern p with respect to t is given by $q_t(p)$.

Let $t_0 = \frac{|\{i \in I \mid t(i) = 1\}|}{|I|}$ be the share of the individuals that fulfill a target t with respect to all individuals and choose $a \in \mathbb{R}$ such that $(i_p)^a$ is well defined. For example, the quality of a k -pattern p with respect to t can now be determined by the following k -quality function.

$$q_t^a(p) = (i_p)^a \cdot (t_p - t_0). \quad (6)$$

Here, we target situations, for example, where t_0 is rather small in comparison to t_p and t_p itself should be much larger. The size of the pattern in terms of contained instances is then weighted by parameter a . Based on this approach, several well-established quality functions can be found in the literature Atzmueller (2015), as shown below:

- The *gain quality function* q_t^0 ,

- The *binomial test quality function* $q_t^{0.5}$,
- The *Piatetsky–Shapiro quality function* q_t^1 .

Then, the goal is to find patterns with a high quality value, i.e., a high $q_t^a(p)$ for a pattern p , with a given a and the target t as the concept of interest.

In the remaining part of this section, strategies are discussed to build and then to evaluate simplicial-based patterns. For a sample data set, we investigate the problem whether simplicial features help to increase the quality of corresponding patterns in comparison to graph-based features. For this purpose, we focus on patterns in the context of specifically constructed synthetic data, which specifically features analysis options for our evaluation strategies.

For generating synthetic data in our application context, we rely on standard approaches from the field of complex networks, applying the susceptible infectious (SI) model, e.g., (Crepey et al. 2006; Li 2018), for generating the non-attacker/attacker structures.

Thus, at first, we create synthetic data by building a simplicial complex, whose vertices are assigned as attackers or non-attackers (which are partially attacked). Using the obtained synthetic dataset, we create features using the available metrics on simplicial complexes and networks, respectively. Using these features, we can then construct patterns with respect to the targets attacker/non-attacker, for studying the impact of metrics on simplicial complexes.

Remark 5.6 The strategy for constructing the synthetic data is given through the following steps.

- (i) Choose an existing and real world social network which has not too many individuals. Then, these correspond to vertices, which are connected by an edge if there is interaction on the level of individuals.
- (ii) Choose a number $k \in \mathbb{N}$ and select k random vertices to be attackers.
- (iii) Create s time periods of attacks on the given social network. Individuals who have been attacked may mutate into new attackers.
- (iv) Determine a resulting network from (iii) for further investigations.
- (v) Build the Vietoris–Rips complex of the underlying network from (iv) using the metric d_5 from Section 2.

In the following we apply this algorithm on one specific social network to illustrate our modeling and analysis approach using simplicial complexes. In particular, for step (i), here we use the congress network from Fink (2023), which is reduced to the first 20 individuals. Thus, all other vertices (with numbers 21–475), as well as all edges involving at least one of such vertices are deleted. The resulting network is illustrated in Fig. 4.

For step (ii), four data points are randomly chosen with respect to the discrete uniform distribution. Then, we perform steps (iii) and (iv) for $s = 1$ and use the *SI model* approach described below. Finally, the Vietoris–Rips distance is chosen as $r = 1/2$ for step (v); see Definition 2.4.

For (iii) it remains to discuss shortly the SI model for modelling disease infections at a time t , which is well-known in the literature (see, e.g., (Li 2018)). Let

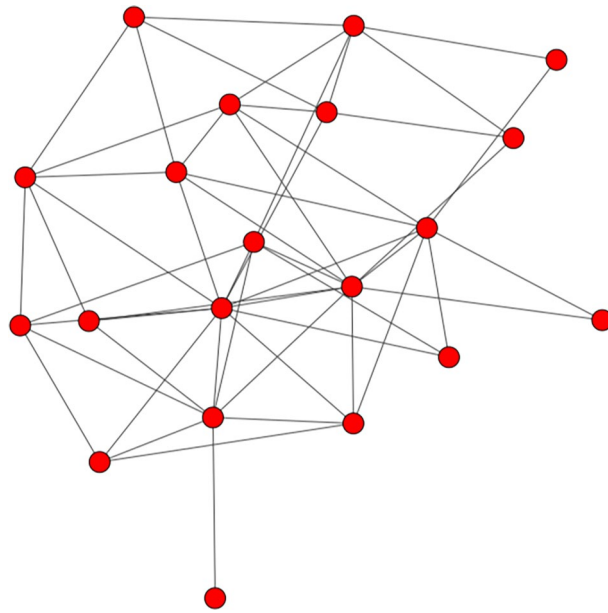


Fig. 4 Modified congress network

S denote the number of susceptible individuals (non-attackers), and
 V the number of virus infectious individuals (attackers),

where

$$S = f(t) \text{ and } V = g(t)$$

for functions f and g with values in $\mathbb{N}$ at a time $t \geq 0$.

There are the following necessary assumptions for the SI model:

- (i) No births and deaths happen, i.e., the number of total individuals N is constant at any time t . This means

$$f(t) + g(t) = N \text{ for all } t \geq 0. \quad (7)$$

- (ii) As suggested in Li (2018, Equation 1.21), we choose the infection reproducing rate as $r = P \cdot \lambda / N$,

where P is the probability of a contact to produce an infection and λ is the average contact number which is also given by the average $(1, 1)$ -degree of the vertices in the corresponding network.

The change of the susceptibles is modeled by a multiple of the product of the number of the two groups (attackers and non-attackers), since this model assumes that the rate of change is proportional to the number of susceptibles and the number of infectious individuals. Moreover, $-\frac{dS}{dt} = \frac{dV}{dt}$ due to Eq. (7).

Hence, the change of infectious and susceptible individuals is described by

$$\frac{dS}{dt} = -cV \cdot S \text{ and } \frac{dV}{dt} = cV \cdot S$$

for a constant $c \in \mathbb{R}$. Thus, one has

$$\frac{dV}{dt} = cNV \cdot (1 - V/N).$$

Assuming that $V \ll N$ it follows $\frac{V}{N} \sim 0$. Hence,

$$\frac{dV}{dt} \approx cN \cdot V = rV$$

with the *change rate* $r = cN$. Since the resulting differential equation

$$\frac{dV}{dt} = rV(1 - \frac{V}{N}) \quad (8)$$

is separable, one obtains the integral equation

$$\int \frac{1}{V(1 - \frac{V}{N})} dV = \int r dt,$$

which can be solved by using partial fraction decomposition.

$$\int r dt = \int \frac{A}{V} + \frac{B}{1 - \frac{V}{N}} dV = \int \frac{A(1 - \frac{V}{N})}{V(1 - \frac{V}{N})} + \frac{BV}{V(1 - \frac{V}{N})} dV$$

leads to the solution $A = 1$ and $B = 1/N$. So, we obtain

$$\int r dt = \int \frac{1}{N(1 - \frac{V}{N})} dV + \int \frac{1}{V} dV.$$

Observe that by calculating the integrals we have

$$rt + c_1 = \ln(V) - \ln(N - V) = \ln \frac{V}{N - V}.$$

Finally, one receives

$$e^{rt} e^{c_1} = \frac{V}{N - V}.$$

This equation can be transformed into

$$V = e^{rt} c_2 N - e^{rt} c_2 V$$

by defining $c_2 := e^{c_1}$. Then, V can be rewritten as

$$V = \frac{e^{rt} c_2 N}{1 + e^{rt} c_2} = \frac{e^{rt} c_2 N}{e^{rt} c_2 (1 + e^{-rt} c_2^{-1})} = \frac{N}{1 + e^{-rt} c} \quad (9)$$

by defining another constant $c := c_2^{-1}$. This constant c can be computed by inserting $t = 0$ into the previous equation. The equation

$$g(0) = \frac{N}{1 + e^{-r \cdot 0} c}$$

leads to

$$c = \frac{N}{g(0)} - 1.$$

Equation (9) implies that

$$g(t) = \frac{N}{1 + (\frac{N}{g(0)} - 1)e^{-rt}}. \quad (10)$$

An analogous consideration for S shows that

$$f(t) = \frac{N}{1 + (\frac{N}{g(0)} - 1)e^{rt}}. \quad (11)$$

In the limit one has

$$\lim_{t \rightarrow \infty} g(t) = \lim_{t \rightarrow \infty} \frac{N}{1 + (\frac{N}{g(0)} - 1) \cdot e^{-rt}} = N \text{ and } \lim_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} \frac{N}{1 + (\frac{N}{g(0)} - 1) \cdot e^{rt}} = 0,$$

since $e^{-rt} \xrightarrow{t \rightarrow \infty} 0$. This means that in the limit everyone is infected.

Applying the SI model and, in particular, Eqs. (10) and (11) we employ the following algorithm for step (iii) in Remark 5.6 using the modified congress data set.

- Algorithm 5.7 (i) Choose randomly a start population of $k = g(0) = 4$ attackers.
(ii) Compute the number of virus infectious individuals $g(1)$ where the probability of a contact to produce an infection is chosen as $P = 0.2$.
(iii) From the set of susceptible individuals, which are connected to at least one infectious individual, choose randomly $g(1) - g(0)$ many individuals with respect to the discrete uniform distribution.
(iv) The set of virus infectious individuals at time 1 is then given by the infectious individual at time 0 together with the new ones.

After applying Algorithm 5.7 on the modified congress network data set, one obtains the attacker data set which is shown in Fig. 5 below (the blue vertices correspond to the attackers).

With the given synthetic data, one is able to build simplicial patterns using measures from Section 4. To keep the discussion simple, we focus on patterns of length 1, which means that we only concentrate on one feature in each pattern.

More precisely, the following features are discussed, where $\mathbb{1}$ is the indicator-function and $k, l \in \mathbb{R}$.

$$(i) \quad \mathbb{1}(c_{\sigma}^{D^{(p,p)}} < k) \text{ and } \mathbb{1}(c_{\sigma}^{D^{(p,p)}} > l),$$

(ii)

$$\mathbb{1}(c_{\sigma}^{E^p} < k) \text{ and } \mathbb{1}(c_{\sigma}^{E^p} > l),$$

$$(iii) \quad \mathbb{1}(c_{\sigma}^{C^p} < k) \text{ and } \mathbb{1}(c_{\sigma}^{C^p} > l).$$

We compare the chosen feature patterns of length 1 for $p = 1$, which corresponds to the graph-based case that is already discussed in Atzmueller et al. (2023), and for larger p with each other by computing their quality $q_{t_i}^0$ with respect to the target of finding

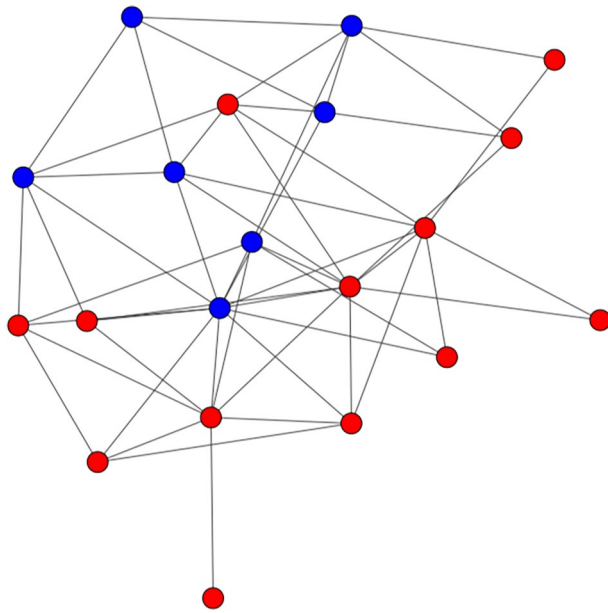


Fig. 5 Modified congress network with attackers after one time step of SI

attackers and non-attackers. This means that targets t_1 and t_2 are considered, where t_1 maps an individual to 1 if it is a non-attacker and otherwise to 0, and t_2 maps an individual to 1 if it is an attacker and otherwise to 0.

For (i) the quality values of the features

$$p_1 = (c_{\sigma}^{D(1,1)} < k_1), p_2 = (c_{\sigma}^{D(2,2)} < k_2) \text{ and } p_3 = (c_{\sigma}^{D(3,3)} < k_3)$$

are analyzed with respect to t_1 . Moreover, the quality values of the features

$$p_3 = (c_{\sigma}^{D(1,1)} > l_1), p_4 = (c_{\sigma}^{D(2,2)} > l_2) \text{ and } p_5 = (c_{\sigma}^{D(3,3)} > l_3)$$

are analyzed with respect to t_2 .

One could also construct length-one patterns using features $c_{\sigma}^{D(p,p)}$ for $p \geq 4$, but in this example it suffices to consider $p < 4$. As illustrated in Fig. 5 no simplices exist in the network with dimensions higher than 3 (see Table 1).

Here we choose

$$k_1 = 4, k_2 = 2 \text{ and } k_3 = 1$$

as well as

$$l_1 = 5, l_2 = 14 \text{ and } l_3 = 7.$$

These choices turn out to deliver the best quality values $q_{t_1}^0$ (resp. $q_{t_2}^0$) for $c_{\sigma}^{D(i,i)} > l_i$ (resp. $c_{\sigma}^{D(i,i)} < k_i$) compared to other possible numbers.

The features as well as the property of being an attacker or a non-attacker are shown in Table 1.

More precisely, the considered patterns have the quality values which are determined via

Table 1 (p, p)-degree centralities

Vertex	$c_D^{(1,1)}$	$c_D^{(2,2)}$	$c_D^{(3,3)}$	$c_D^{(4,4)}$	Attacker?	Non-attacker?
0	6	5	1	0	0	1
1	3	2	0	0	0	1
2	3	1	0	0	0	1
3	5	6	2	0	0	1
4	6	5	0	0	1	0
5	2	1	0	0	0	1
6	1	0	0	0	0	1
7	5	7	3	0	0	1
8	8	12	6	0	0	1
9	6	3	0	0	1	0
10	2	0	0	0	0	1
11	8	9	3	0	0	1
12	6	8	2	0	1	0
13	10	14	6	0	0	1
14	4	2	0	0	1	0
15	6	6	2	0	1	0
16	4	4	1	0	0	1
17	12	21	8	0	1	0
18	5	3	0	0	1	0
19	4	5	2	0	0	1
Vertex	p_1	p_2	p_3	p_4	p_5	p_6
0	0	0	0	1	0	0
1	1	0	1	0	0	0
2	1	1	1	0	0	0
3	0	0	0	0	0	0
4	0	0	1	1	0	0
5	1	1	1	0	0	0
6	1	1	1	0	0	0
7	0	0	0	0	0	0
8	0	0	0	1	0	0
9	0	0	1	1	0	0
10	1	1	1	0	0	0
11	0	0	0	1	0	0
12	0	0	0	1	0	0
13	0	0	0	1	0	0
14	0	0	1	0	0	0
15	0	0	0	1	0	0
16	0	0	0	0	0	0
17	0	0	0	1	1	1
18	0	0	1	0	0	0
19	0	0	0	0	0	0

$$\begin{aligned}
 q_{t_1}^0(p_1) &= i_p^0 \cdot (t_p - t_0) = 5^0 \cdot (5/5 - 13/20) = 7/20, \\
 q_{t_1}^0(p_2) &= 4^0 \cdot (4/4 - 13/20) = 7/20, \\
 q_{t_1}^0(p_3) &= 9^0 \cdot (5/9 - 13/20) = 17/180, \\
 q_{t_2}^0(p_4) &= 9^0 \cdot (5/9 - 7/20) = 37/180, \\
 q_{t_2}^0(p_5) &= 1^0 \cdot (1 - 7/20) = 13/20, \\
 q_{t_2}^0(p_6) &= 1^0 \cdot (1 - 7/20) = 13/20.
 \end{aligned}$$

Thus, the (1, 1)-degree centrality is as good as the (2, 2)-degree centrality, and they have both a higher quality than the (3, 3)-degree centrality, with respect to the pattern t_1 and the chosen quality function. The (2, 2)-degree centrality is as good as the (3, 3)-degree centrality and they have both a higher quality than the (1, 1)-degree centrality with respect to the pattern t_2 and $q_{t_2}^0$. Hence, $p = 2$ is the best choice for considering a simplicial degree centrality feature pattern of only length 1, given the described setup.

The features in (ii) are considered in Table 2. Here we analyze the following patterns with respect to t_1 .

$$p_1 = (c_{\sigma}^{E^1} < k_1), p_2 = (c_{\sigma}^{E^2} < k_2) \text{ and } p_3 = (c_{\sigma}^{E^3} < k_3)$$

with

$$k_1 = 0.5, k_2 = 0.2 \text{ and } k_3 = 2.$$

Moreover, we investigate the following patterns with respect to t_2 .

$$p_4 = (c_{\sigma}^{E^1} > l_1), p_5 = (c_{\sigma}^{E^2} > l_2) \text{ and } p_6 = (c_{\sigma}^{E^3} > l_3)$$

with

$$l_1 = 2.5, l_2 = 2.5 \text{ and } l_3 = 2.5.$$

Table 2 p -eigenvector centralities

Vertex	$c(E^1)$	$c(E^2)$	$c(E^3)$	$c(E^4)$	Attacker?	Non-attacker?	p_1	p_2	p_3	p_4	p_5	p_6
0	1.421	1.279	0.913	0.0	0	1	0	0	1	0	1	0
1	0.962	0.973	0.0	0.0	0	1	0	0	1	0	0	0
2	0.663	0.122	0.0	0.0	0	1	0	1	1	0	0	0
3	1.384	1.430	1.338	0.0	0	1	0	0	1	0	0	0
4	1.442	1.455	0.0	0.0	1	0	0	0	1	0	0	0
5	0.648	0.661	0.0	0.0	0	1	0	0	1	0	0	0
6	0.298	0.0	0.0	0.0	0	1	1	1	1	0	0	0
7	1.564	1.617	1.760	0.0	0	1	0	0	1	0	0	0
8	1.935	1.962	2.134	0.0	0	1	0	0	0	0	0	0
9	0.908	0.384	0.0	0.0	1	0	0	0	1	0	0	0
10	0.421	0.0	0.0	0.0	0	1	1	1	1	0	0	0
11	1.828	1.803	1.564	0.0	0	1	0	0	1	0	0	0
12	1.643	1.640	1.299	0.0	1	0	0	0	1	0	0	0
13	2.380	2.336	2.293	0.0	0	1	0	0	0	0	0	0
14	0.772	0.617	0.0	0.0	1	0	0	0	1	0	0	0
15	1.601	1.514	1.475	0.0	1	0	0	0	1	0	0	0
16	1.321	1.362	1.238	0.0	0	1	0	0	1	0	0	0
17	2.816	2.772	2.565	0.0	1	0	0	0	0	1	1	1
18	1.014	0.384	0.0	0.0	1	0	0	0	1	0	0	0
19	1.187	1.244	1.381	0.0	0	1	0	0	1	0	0	0

Table 3 p -closeness centralities

Vertex	Attacker?	Non-attacker?	c^{C^1}	p_1	p_2
0	0	1	0.029	1	0
1	1	0	0.026	0	0
2	1	0	0.025	0	0
3	0	1	0.028	1	0
4	0	1	0.028	1	0
5	1	0	0.023	0	1
6	1	0	0.02	0	1
7	1	0	0.029	1	0
8	1	0	0.031	1	0
9	1	0	0.027	1	0
10	1	0	0.023	0	1
11	1	0	0.032	1	0
12	0	1	0.030	1	0
13	1	0	0.036	1	0
14	1	0	0.024	0	0
15	0	1	0.031	1	0
16	1	0	0.028	1	0
17	1	0	0.039	1	0
18	0	1	0.029	1	0
19	1	0	0.025	1	0

Here the chosen k_1, k_2, k_3 and l_1, l_2, l_3 deliver the best quality values with respect to $q_{t_1}^0$ and $q_{t_2}^0$. These numbers are not unique with this property. e.g., for l_3 one can also choose 2.4.

It is not helpful to consider eigenvector centralities for $p \geq 4$ for the modified congress network, since then the eigenvector centralities are 0 for all vertices (see Table 2 for examples). Thus, it is not possible to use such $p \geq 4$ to create patterns for distinguishing attackers and non-attackers.

A computation yields

$$\begin{aligned}
 q_{t_1}^0(p_1) &= 2^0 \cdot (2/2 - 13/20) = 7/20, \\
 q_{t_1}^0(p_2) &= 3^0 \cdot (3/3 - 13/20) = 7/20, \\
 q_{t_1}^0(p_3) &= 17^0 \cdot (12/17 - 13/20) = 19/340, \\
 q_{t_2}^0(p_4) &= 1^0 \cdot (1/1 - 7/20) = 13/20, \\
 q_{t_2}^0(p_5) &= 1^0 \cdot (1/1 - 7/20) = 13/20, \\
 q_{t_2}^0(p_6) &= 1^0 \cdot (1/1 - 7/20) = 13/20.
 \end{aligned}$$

Thus, the 1-eigenvector centrality is as good as the 2-eigenvector centrality, and they have both a higher quality than the 3-eigenvector centrality with respect to the pattern t_1 and the chosen quality function. The 1-eigenvector centrality is as useful as the 2-eigenvector centrality and the 3-eigenvector centrality with respect to the pattern t_2 . So, in total, the eigenvector-centralities for $p = 1$ and $p = 2$ are the best choices regarding the simplicial-eigenvector-feature-patterns of length 1.

For (iii) only features for $c_\sigma^{C^1}$ are considered, since the closeness centrality measure is always 0 for other p , see Table 3. The best quality for characterizing non-attackers (target t_1) is achieved with the pattern

$$p_1 = (c_\sigma^{C^1} < 0.024).$$

Moreover, the feature

$$p_2 = (c_{\sigma}^{G^1} > 0.027)$$

has the highest quality with respect to target t_2 , i.e., for characterizing attackers. As in (ii) the chosen numbers are not unique with respect to this property.

More precisely, p_1 and p_2 have the following quality values.

$$\begin{aligned} q_{t_1}^0(p_1) &= i_p \cdot (t_p - t_0) = 3^0 \cdot (3/3 - 13/20) = 7/20, \\ q_{t_2}^0(p_2) &= i_p \cdot (t_p - t_0) = 13^0 \cdot (6/13 - 7/20) = 29/260. \end{aligned}$$

The pattern p_1 has a rather high quality and is thus useful for further investigations. The constructed feature pattern with respect to the attacker target t_2 is not that beneficial, but there may be additional other possibilities for using this feature (which are not considered here in this manuscript), like combining it with other features in longer patterns. Observe that, in contrast to (i) and (ii), in (iii) for the modified congress network, it was not possible to construct helpful simplicial features of complexes of higher dimensions than 1, but only graph-based features using the closeness centrality. In consequence, it would be reasonable to consider here in the future other networks, where there might be a denser structure for higher dimensional simplices to find vertices that have non-zero p -closeness centrality for $p > 1$.

Remark 5.8 To avoid the synthetic data used in this section one has to record non-trivial real intrusion data, which contains not only the attacks, but also the interaction in-between groups of attackers and of attacked IP-addresses. This is left as an future research problem.

Outlook

In this work we applied simplicial complexes for detecting higher-order patterns as simplicial-based patterns, in the context of network intrusion detection settings. In Section 5 we considered a synthetic data set and assessed how good the considered simplicial feature patterns worked for it.

In the following we outline perspectives on further eligible strategies to evaluate the usage of simplicial-based patterns for describing targets, e.g., to differentiate attacker and non-attackers in networks based on real data. There are several options to continue the investigations from our work, including the following.

- (i) Using various (further) datasets to analyze the utility of simplicial feature patterns. For this it is reasonable to vary synthetic or real large or small data as well as data from different application domains.
- (ii) Constructing patterns that use more simplicial complex features at once. It is reasonable to consider patterns that have length > 1 to study further centrality measures which were not yet mentioned in this manuscript.
- (iii) Equipping simplicial patterns with further features which are not originating from simplicial centrality measures. It is also possible to construct further features from one given centrality measure by using new conditions. In the following we provide some examples for the latter suggestion.

- The condition that the maximal dimension of a vertex- v -including facet is higher than k , can be described as

$$\mathbb{1}\left(\max_{v \in \sigma, \sigma \in \Delta} \dim(\sigma) > k\right).$$

- The condition, that the importance of a vertex- v -including facet in the simplicial complex regarding the centrality measure c is higher than k , can be described as

$$\mathbb{1}\left(\max_{v \in \sigma, \sigma \in \Delta} c_{\sigma} > k\right).$$

- The condition, that the maximal eigenvector-centrality of a vertex σ is higher than k , can be described as

$$\mathbb{1}(c_{\sigma}^{C^*} > k).$$

- (iv) Using multiple types of simplicial complexes, like Čech complex and Vietoris–Rips complex, and combine features depending on several simplicial complexes in one pattern. Consider different metrics and select complexes that lead to the best results.

Acknowledgements

This work was supported by the Lower Saxony Ministry of Science and Culture (MWK), through the zukunft.niedersachsen program of the Volkswagen Foundation ("Joint-Lab Artificial Intelligence & Data Science"), and by the German Research Foundation (DFG), grant MODUS-II (316679917, AT 88/4-2).

Author contributions

MW, MA and TR contributed in conceptualizing the idea, discussing the methodology of the paper, as well as design the experimentation. MW drafted the initial version of the paper in discussion with MA and TR. All authors reviewed, revised and edited, and also approved the final manuscript.

Funding

Open Access funding enabled and organized by Projekt DEAL. This work was supported by the Lower Saxony Ministry of Science and Culture (MWK), through the zukunft.niedersachsen program of the Volkswagen Foundation ("Joint-Lab Artificial Intelligence & Data Science"), and by the German Research Foundation (DFG), grant MODUS-II (316679917, AT 88/4-2).

Data availability

The investigated data was manually build as described in Section 5 from the Twitter dataset from Fink (2023). One can download the original dataset from the URL <https://snap.stanford.edu/data/higgs-twitter.html>.

Declarations

Competing interests

The authors are not aware of any conflict of interest.

Author details

¹Applied Algebra and Data Analysis Group, Osnabrück University, Albrechtstr. 28a, 49076 Osnabrück, Germany

²Semantic Information Systems Group, Osnabrück University, Wachsbleiche 27, 49090 Osnabrück, Germany

³Research Department Cooperative and Autonomous Systems, German Research Center for Artificial Intelligence (DFKI), Hamburger Str. 24, 49084 Osnabrück, Germany

⁴Joint Lab Künstliche Intelligenz & Data Science (JL KI&DS), Hamburger Str. 24, 49084 Osnabrück, Germany

Received: 6 December 2024 / Accepted: 16 June 2025

Published online: 23 July 2025

References

- Akinwande G, Reitzner M (2020) Multivariate central limit theorems for random simplicial complexes. *Adv Appl Math* 121:102076
- Atzmueller M (2015) Subgroup discovery. *WIREs Data Min Knowl Discovery* 5:35–49
- Atzmueller M, Soldano H, Santini G, Bouthinon D (2019) MinerLSD: efficient mining of local patterns on attributed networks. *Appl Netw Sci* 4:1–33
- Atzmueller M, Sylvester S, Kanawati R (2023) Exploratory and explanation-aware network intrusion profiling using subgroup discovery and complex network analysis. In: *ACM international conference proceeding series* pp 153–158

- Bobrowski O, Kahle M (2018) Topology of random geometric complexes: a survey. *J Appl Comput Topol* 1:331–364
- Borgatti SP, Everett MG (2006) A graph-theoretic perspective on centrality. *Soc Netw* 28:466–484
- Crepey P, Alvarez FP, Barthélemy M (2006) Epidemic variability in complex networks. *Phys Rev E* 73:046131
- Diestel R (2025) *Graph Theory*, 6 edn Springer
- Edelsbrunner H, Harer JL (2022) *Computational topology: an introduction*, American Mathematical Society
- Fink CG et al (2023) A centrality measure for quantifying spread on weighted, directed networks. *Physica A* 626:129083
- Gaubert S, Gunawardena J (2004) The Perron-Frobenius theorem for homogeneous, monotone functions. *Trans Am Math Soc* 356:4931–4950
- Gilbert EN (1961) Random plane networks. *J Soc Indus Appl Math* 9:533–543
- Grosskreutz H, Rüping S, Wrobel S (2008) Tight optimistic estimates for fast subgroup discovery. *Joint European conference on machine learning and knowledge discovery in databases* 440–456
- Grygierek J, Juhnke-Kubitzke M, Reitzner M, Römer T, Røndigs O (2020) Gigantic random simplicial complexes. *Homology Homotopy Appl* 22:297–318
- Hug D, Reitzner M (2016) Introduction to stochastic geometry. *Stochastic analysis for Poisson point processes: Malliavin Calculus, Wiener-Itô chaos expansions and stochastic geometry* 145–184
- Kahle M, Meckes E (2013) Limit theorems for Betti numbers of random simplicial complexes. *Homol Homotopy Appl* 15:343–374
- Kosk R et al (2024) Deep spectral meshes: multi-frequency facial mesh processing with graph neural networks. *arXiv preprint. arXiv:2402.10365*
- Kozlov D (2008) *Combinatorial algebraic topology*, Springer
- Lemmens B, Nussbaum R (2012) *Nonlinear Perron-Frobenius Theory*, vol 189, Cambridge University Press
- Li M (2018) *An introduction to mathematical modeling of infectious diseases*. *Mathematics of Planet Earth*, Springer International Publishing
- Marisca I, Alippi C, Bianchi FM (2024) Graph-based Forecasting with Missing Data through Spatiotemporal Downsampling. *arXiv preprint. arXiv:2402.10634*
- Mukherjee B, Heberlein LT, Levitt KN (1994) Network intrusion detection. *IEEE Netw* 8:26–41
- Munkres JR (2018) *Elements of algebraic topology* (CRC press)
- Newman M (2018) *Networks*, Oxford university press
- Penrose M (2003) *Random geometric graphs*, Vol. 5, Oxford University Press
- Reitzner M, Römer T, von Westenholz M (2024) Covariance matrices of length power functionals of random geometric graphs—an asymptotic analysis. *Linear Algebra Appl* 691:151–181
- Rosenberg J (2017) Security in embedded systems. *Rugged Embedded Syst* 3
- Serrano DH, Gómez DS (2020) Centrality measures in simplicial complexes: applications of topological data analysis to network science. *Appl Math Comput* 382:125331
- Sommer R, Paxson V (2010) Outside the closed world: on using machine learning for network intrusion detection. *2010 IEEE symposium on security and privacy* 305–316
- Stanley RP (2007) *Combinatorics and commutative algebra*, vol 41, Springer
- Tang J et al (2023) Graphgpt: graph instruction tuning for large language models. *arXiv preprint. arXiv:2310.13023*
- Tirumala S, Valluri MR, Babu G (2019) A survey on cybersecurity awareness concerns, practices and conceptual measures. *2019 international conference on computer communication and informatics (ICCCI)* 1–6
- Walters R, Novak M (2021) Cyber security. *Cyber security, artificial intelligence, data protection & the law* 21–37

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.